## Exercises Periodic Table

1. **What are the horizontal lines in the periodic table called? What can we read from them?**
   Periods. Among other things, the period directly indicates the electron shell.

2. **What are the vertical lines called? What can we read from them?**
   Groups. Among other things, the group indicates the number of valence electrons.

3. **Name two metals and two nonmetals.**
   Metals: iron, manganese. Nonmetals: oxygen, nitrogen.

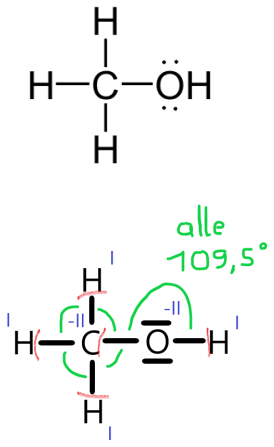
4. **Why do alkali metals react so explosively with water?**
   Their single valence electron is only weakly bound (very low electronegativity), so it is easily donated—this happens on contact with water. Hydrogen gas, base (alkali), and heat are produced; the heat ignites the hydrogen gas (activation energy).

5. **What is the most common reaction of halogens? What happens in it?**
   Halogens are strong oxidizing agents. They readily oxidize other substances such as metals. When a halogen reacts with a metal, a salt is formed in a redox reaction.

6. **Why is water polar?**
   Because hydrogen has a lower electronegativity than oxygen, the H–O bond is polar. Water contains two H–O bonds at an angle of about 109.5°, so this unequal electron distribution is not canceled by symmetry. The molecule has an asymmetric electron distribution overall and therefore a dipole—so water is a polar molecule.

7. **Determine all oxidation numbers and bond angles of the two molecules shown. Do they have a dipole?**




   Both have polar bonds that are not cancelled by symmetry; both molecules therefore have a dipole. (Angles shown: ~180° for linear; ~109.5° for tetrahedral.)

8. **What are the most abundant elements in the human body?**
   Decreasing mass percentages: O, C, H, N, …

9. **What are trace elements?**

Trace elements are essential for the body’s function but occur at less than 50 mg/kg body weight—hence 'trace'. If not supplied in sufficient amounts, they can cause sometimes severe deficiency symptoms; common examples include iron or iodine deficiency. Many trace elements are present in dietary supplements, vitamin preparations, or 'Schüssler salts'. With an unbalanced diet, particular attention must be paid to adequate trace-element intake. Projects like vitamin-enriched Golden Rice address deficiencies in poorer populations.